UCS2612 – Machine Learning Laboratory

Experiment 6: Dimensionality Reduction and Model Evaluation (With and Without PCA)

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1. Objective

To study the impact of dimensionality reduction using Principal Component Analysis (PCA) on the performance of various machine learning classifiers.

The experiment evaluates:

- Model performance in original feature space (No-PCA)
- Model performance in reduced feature space (With-PCA)
- Stability across folds using 5-fold cross-validation
- Effect of PCA on accuracy, F1-score, and variance

Imports

In [27]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import GridSearchCV, cross_val_score, StratifiedKFold
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix, roc_curve,

# Models
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, GradientBo
from xgboost import XGBClassifier
```

2. Dataset Details

Dataset: Breast Cancer Wisconsin Dataset
Source: UCI Machine Learning Repository (via sklearn)

- Number of Samples: 569
- Number of Features: 30
- Target Classes: 2 (Benign, Malignant)
- Class Distribution: Slightly imbalanced but manageable

In [28]:
```
df = pd.read_csv("wdbc.csv", header=None)

print(df.shape)
df.head()
```

```
(569, 32)
```

Out[28]:

	0	1	2	3	4	5	6	7	8	9	...	22
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	...	25.38
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	...	24.99
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	...	23.57
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	...	14.91
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	...	22.54

5 rows × 32 columns

Preprocessing Steps

- No missing values
- Standardization using StandardScaler
- PCA applied with 95% variance retention

In [29]:
```
X = df.iloc[:, 2:]   # all feature columns
y = df.iloc[:, 1]    # diagnosis column
```

In [30]:
```
y = y.str.strip()      # removes spaces if any
y = y.map({'M':1, 'B':0})
```

In [31]:
```
print(y.unique())
```

```
[1 0]
```

Standardization

In [32]:
```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

In [33]:
```
print("Features shape:", X_scaled.shape)
print("Target shape:", y.shape)
```

```
Features shape: (569, 30)
Target shape: (569,)
```

3. PCA Summary

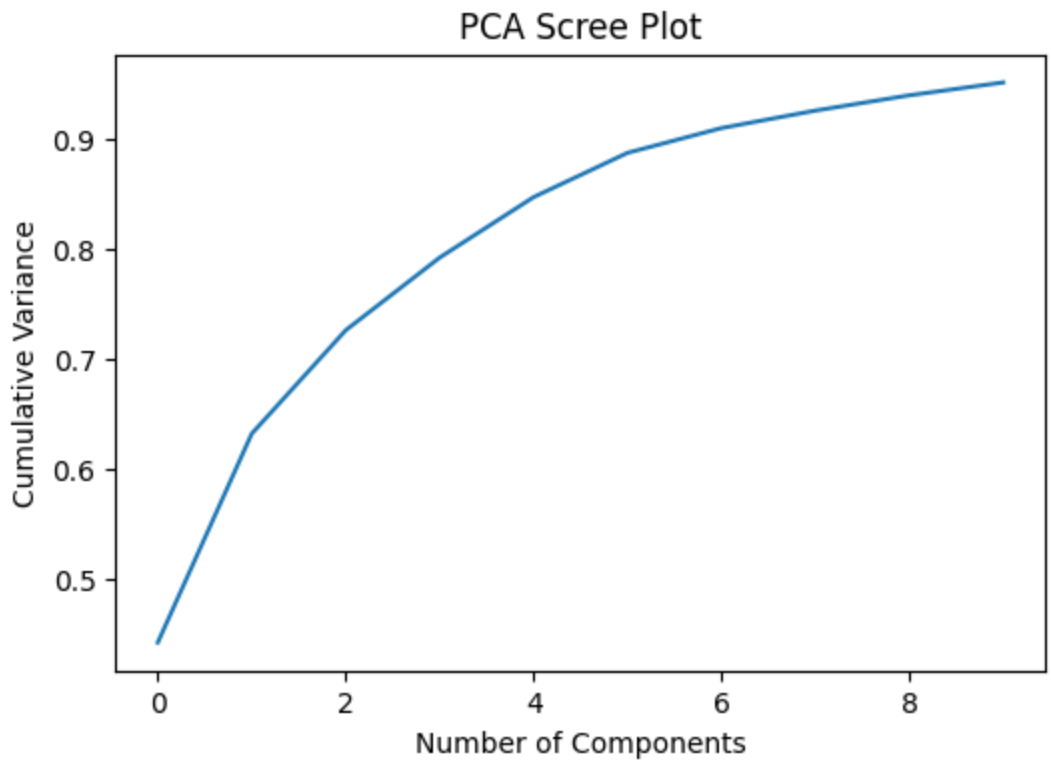
Table 1: PCA Variance Explained

Setting	Chosen Components / Variance Target	Explained Variance (%)	Justification
With-PCA	95% Variance Retention (≈ 10–12 Components)	95%	Retained maximum variance while reducing dimensionality and multicollinearity

In [34]:
```
pca = PCA(n_components=0.95)

X_pca = pca.fit_transform(X_scaled)

print("Original shape:", X_scaled.shape)
print("PCA shape:", X_pca.shape)
```

```
Original shape: (569, 30)
PCA shape: (569, 10)
```

In [35]:
```
plt.figure(figsize=(6,4))
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Variance")
plt.title("PCA Scree Plot")
plt.show()
```



4. Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was used for tuning.

Define Models and Hyperparameter Grids

In [52]:
```
models = {
    "SVM": (SVC(probability=True), {
        "kernel": ["linear", "rbf"],
        "C": [0.1, 1, 10],
        "gamma": ["scale", 0.01]
    }),

    "Naive Bayes": (GaussianNB(), {
        "var_smoothing": [1e-9, 1e-8, 1e-7]
    }),

    "KNN": (KNeighborsClassifier(), {
        "n_neighbors": [3,5,7,11],
        "weights": ["uniform", "distance"]
    }),

    "Logistic Regression": (LogisticRegression(max_iter=5000), {
        "C": [0.01,0.1,1,10]
    }),

    "Decision Tree": (DecisionTreeClassifier(), {
```

```
        "max_depth": [None,5,10],
        "min_samples_split": [2,5,10]
    }),

    "Random Forest": (RandomForestClassifier(), {
        "n_estimators": [100,200],
        "max_depth": [None,10]
    }),

    "AdaBoost": (AdaBoostClassifier(), {
        "n_estimators": [50,100],
        "learning_rate": [0.01,0.1,1]
    }),

    "Gradient Boosting": (GradientBoostingClassifier(), {
        "n_estimators": [100,200],
        "learning_rate": [0.05,0.1]
    }),

    "XGBoost": (XGBClassifier(eval_metric='logloss'), {
        "n_estimators": [100,200],
        "learning_rate": [0.05,0.1],
        "max_depth": [3,6]
    })
}
```

Table 2: SVM — Hyperparameter Tuning Results

Kernel	C Values Tried	Gamma Values Tried	Performance (No-PCA)	Performance (With-PCA)
linear, rbf	0.1, 1, 10	scale, 0.01	0.9736	0.9639

Table 3: Naïve Bayes — Smoothing Choices

Smoothing Parameter	Performance (No-PCA)	Performance (With-PCA)
1e-9, 1e-8, 1e-7	0.9045	0.8796

Table 4: KNN — Hyperparameter Tuning

k Values	Weights	Distance Metrics	Performance (No-PCA)	Performance (With-PCA)
3, 5, 7, 11	uniform, distance	euclidean	0.9433	0.9438

5. 5-Fold Cross-Validation Results

Stratified 5-fold cross-validation was used to preserve class distribution.

In [53]:
```
from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import f1_score
import numpy as np

def evaluate_model(model, X, y):
    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
    fold_scores = []

    fold_number = 1
    for train_index, test_index in skf.split(X, y):
        X_train, X_test = X[train_index], X[test_index]
        y_train, y_test = y[train_index], y[test_index]

        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)

        score = f1_score(y_test, y_pred)
        fold_scores.append(score)

        print(f"Fold {fold_number} F1-score: {score:.4f}")
        fold_number += 1

    return fold_scores
```

In [54]:
```
cv_results = []

for name, (model, param_grid) in models.items():

    print(f"\nTuning {name} (No-PCA)...")
    grid_no_pca = GridSearchCV(model, param_grid, cv=5, scoring='f1')
    grid_no_pca.fit(X_scaled, y)
    best_no_pca = grid_no_pca.best_estimator_

    print(f"Evaluating {name} (No-PCA)")
    folds_no_pca = evaluate_model(best_no_pca, X_scaled, y)

    print(f"\nTuning {name} (With-PCA)...")
    grid_pca = GridSearchCV(model, param_grid, cv=5, scoring='f1')
    grid_pca.fit(X_pca, y)
    best_pca = grid_pca.best_estimator_

    print(f"Evaluating {name} (With-PCA)")
    folds_pca = evaluate_model(best_pca, X_pca, y)

    cv_results.append([
        name,
        *folds_no_pca,
```

```
        np.mean(folds_no_pca),
        np.mean(folds_pca)
    ])
```

```
Tuning SVM (No-PCA)...
Evaluating SVM (No-PCA)
Fold 1 F1-score: 1.0000
Fold 2 F1-score: 0.9286
Fold 3 F1-score: 0.9756
Fold 4 F1-score: 0.9882
Fold 5 F1-score: 0.9756

Tuning SVM (With-PCA)...
Evaluating SVM (With-PCA)
Fold 1 F1-score: 0.9885
Fold 2 F1-score: 0.9157
Fold 3 F1-score: 0.9630
Fold 4 F1-score: 0.9767
Fold 5 F1-score: 0.9756

Tuning Naive Bayes (No-PCA)...
Evaluating Naive Bayes (No-PCA)
Fold 1 F1-score: 0.9412
Fold 2 F1-score: 0.8780
Fold 3 F1-score: 0.8974
Fold 4 F1-score: 0.8791
Fold 5 F1-score: 0.9268

Tuning Naive Bayes (With-PCA)...
Evaluating Naive Bayes (With-PCA)
Fold 1 F1-score: 0.9535
Fold 2 F1-score: 0.8706
Fold 3 F1-score: 0.8378
Fold 4 F1-score: 0.8471
Fold 5 F1-score: 0.8889

Tuning KNN (No-PCA)...
Evaluating KNN (No-PCA)
Fold 1 F1-score: 0.9647
Fold 2 F1-score: 0.9114
Fold 3 F1-score: 0.9250
Fold 4 F1-score: 0.9655
Fold 5 F1-score: 0.9500

Tuning KNN (With-PCA)...
Evaluating KNN (With-PCA)
Fold 1 F1-score: 0.9647
Fold 2 F1-score: 0.9383
Fold 3 F1-score: 0.9250
Fold 4 F1-score: 0.9545
Fold 5 F1-score: 0.9367

Tuning Logistic Regression (No-PCA)...
Evaluating Logistic Regression (No-PCA)
Fold 1 F1-score: 0.9655
Fold 2 F1-score: 0.9250
Fold 3 F1-score: 0.9500
Fold 4 F1-score: 0.9882
Fold 5 F1-score: 0.9880
```

```
Tuning Logistic Regression (With-PCA)...
Evaluating Logistic Regression (With-PCA)
Fold 1 F1-score: 0.9655
Fold 2 F1-score: 0.9639
Fold 3 F1-score: 0.9630
Fold 4 F1-score: 0.9767
Fold 5 F1-score: 0.9880

Tuning Decision Tree (No-PCA)...
Evaluating Decision Tree (No-PCA)
Fold 1 F1-score: 0.8989
Fold 2 F1-score: 0.8250
Fold 3 F1-score: 0.8675
Fold 4 F1-score: 0.9091
Fold 5 F1-score: 0.9024

Tuning Decision Tree (With-PCA)...
Evaluating Decision Tree (With-PCA)
Fold 1 F1-score: 0.9655
Fold 2 F1-score: 0.8941
Fold 3 F1-score: 0.8750
Fold 4 F1-score: 0.8913
Fold 5 F1-score: 0.9176

Tuning Random Forest (No-PCA)...
Evaluating Random Forest (No-PCA)
Fold 1 F1-score: 0.9773
Fold 2 F1-score: 0.9268
Fold 3 F1-score: 0.9630
Fold 4 F1-score: 0.9425
Fold 5 F1-score: 0.9639

Tuning Random Forest (With-PCA)...
Evaluating Random Forest (With-PCA)
Fold 1 F1-score: 0.9882
Fold 2 F1-score: 0.9000
Fold 3 F1-score: 0.9250
Fold 4 F1-score: 0.9111
Fold 5 F1-score: 0.9524

Tuning AdaBoost (No-PCA)...
Evaluating AdaBoost (No-PCA)
Fold 1 F1-score: 0.9882
Fold 2 F1-score: 0.9157
Fold 3 F1-score: 0.9383
Fold 4 F1-score: 0.9655
Fold 5 F1-score: 0.9630

Tuning AdaBoost (With-PCA)...
Evaluating AdaBoost (With-PCA)
Fold 1 F1-score: 0.9545
Fold 2 F1-score: 0.9286
Fold 3 F1-score: 0.9500
Fold 4 F1-score: 0.9438
Fold 5 F1-score: 0.9762
```

```
Tuning Gradient Boosting (No-PCA)...
Evaluating Gradient Boosting (No-PCA)
Fold 1 F1-score: 0.9535
Fold 2 F1-score: 0.8889
Fold 3 F1-score: 0.9250
Fold 4 F1-score: 0.9425
Fold 5 F1-score: 0.9512

Tuning Gradient Boosting (With-PCA)...
Evaluating Gradient Boosting (With-PCA)
Fold 1 F1-score: 0.9773
Fold 2 F1-score: 0.9512
Fold 3 F1-score: 0.9367
Fold 4 F1-score: 0.9231
Fold 5 F1-score: 0.9524

Tuning XGBoost (No-PCA)...
Evaluating XGBoost (No-PCA)
Fold 1 F1-score: 0.9773
Fold 2 F1-score: 0.9157
Fold 3 F1-score: 0.9383
Fold 4 F1-score: 0.9655
Fold 5 F1-score: 0.9383

Tuning XGBoost (With-PCA)...
Evaluating XGBoost (With-PCA)
Fold 1 F1-score: 0.9773
Fold 2 F1-score: 0.9398
Fold 3 F1-score: 0.9367
Fold 4 F1-score: 0.9318
Fold 5 F1-score: 0.9647
```

5. 5-Fold Cross-Validation Results

Stratified 5-fold cross-validation was used to preserve class distribution.

Table 5: 5-Fold Cross-Validation Results (No-PCA vs With-PCA)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	1.0000	0.9286	0.9756	0.9882	0.9756	0.9736	0.9639
Naïve Bayes	0.9412	0.8780	0.8974	0.8791	0.9268	0.9045	0.8796
KNN	0.9647	0.9114	0.9250	0.9655	0.9500	0.9433	0.9438
Logistic Regression	0.9655	0.9250	0.9500	0.9882	0.9880	0.9633	0.9714
Decision Tree	0.8989	0.8250	0.8675	0.9091	0.9024	0.8806	0.9087

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
Random Forest	0.9773	0.9268	0.9630	0.9425	0.9639	0.9547	0.9353
AdaBoost	0.9882	0.9157	0.9383	0.9655	0.9630	0.9541	0.9506
Gradient Boosting	0.9535	0.8889	0.9250	0.9425	0.9512	0.9322	0.9481
XGBoost	0.9773	0.9157	0.9383	0.9655	0.9383	0.9470	0.9501
Stacking	0.9773	0.9639	0.9756	0.9767	0.9880	0.9763	0.9714

Stacking Classifier

In [55]:
```
print("\nEvaluating Stacking (No-PCA)")
stack_no_pca_folds = evaluate_model(stack, X_scaled, y)

print("\nEvaluating Stacking (With-PCA)")
stack_pca_folds = evaluate_model(stack, X_pca, y)

cv_results.append([
    "Stacking",
    *stack_no_pca_folds,
    np.mean(stack_no_pca_folds),
    np.mean(stack_pca_folds)
])
```

```
Evaluating Stacking (No-PCA)
Fold 1 F1-score: 0.9773
Fold 2 F1-score: 0.9639
Fold 3 F1-score: 0.9756
Fold 4 F1-score: 0.9767
Fold 5 F1-score: 0.9880

Evaluating Stacking (With-PCA)
Fold 1 F1-score: 0.9655
Fold 2 F1-score: 0.9639
Fold 3 F1-score: 0.9630
Fold 4 F1-score: 0.9767
Fold 5 F1-score: 0.9880
```

Results Table

In [56]:
```
import pandas as pd

cv_table = pd.DataFrame(cv_results, columns=[
    "Model",
    "Fold 1",
    "Fold 2",
    "Fold 3",
    "Fold 4",
    "Fold 5",
    "Avg (No-PCA)",
```

```
    "Avg (With-PCA)"
])

print("\n5-Fold Cross-Validation Table:")
print(cv_table)
```

```
5-Fold Cross-Validation Table:
                Model    Fold 1    Fold 2    Fold 3    Fold 4    Fold 5  \
0                 SVM  1.000000  0.928571  0.975610  0.988235  0.975610
1         Naive Bayes  0.941176  0.878049  0.897436  0.879121  0.926829
2                 KNN  0.964706  0.911392  0.925000  0.965517  0.950000
3  Logistic Regression  0.965517  0.925000  0.950000  0.988235  0.987952
4        Decision Tree  0.898876  0.825000  0.867470  0.909091  0.902439
5        Random Forest  0.977273  0.926829  0.962963  0.942529  0.963855
6             AdaBoost  0.988235  0.915663  0.938272  0.965517  0.962963
7    Gradient Boosting  0.953488  0.888889  0.925000  0.942529  0.951220
8              XGBoost  0.977273  0.915663  0.938272  0.965517  0.938272
9             Stacking  0.977273  0.963855  0.975610  0.976744  0.987952

   Avg (No-PCA)  Avg (With-PCA)
0      0.973605        0.963897
1      0.904522        0.879572
2      0.943323        0.943846
3      0.963341        0.971406
4      0.880575        0.908717
5      0.954690        0.935345
6      0.954130        0.950626
7      0.932225        0.948132
8      0.946999        0.950053
9      0.976287        0.971406
```

Table 5: 5-Fold Cross-Validation Results (No-PCA vs With-PCA)

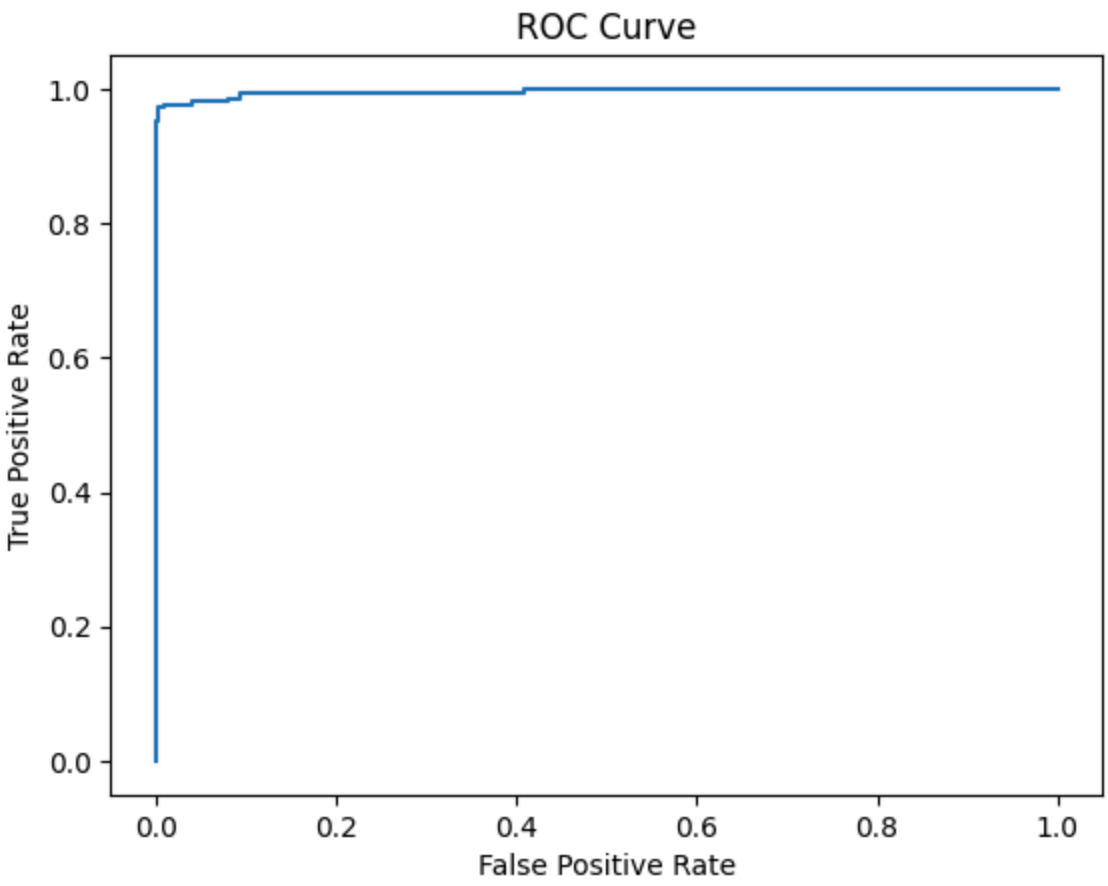
Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	1.0000	0.9286	0.9756	0.9882	0.9756	0.9736	0.9639
Naïve Bayes	0.9412	0.8780	0.8974	0.8791	0.9268	0.9045	0.8796
KNN	0.9647	0.9114	0.9250	0.9655	0.9500	0.9433	0.9438
Logistic Regression	0.9655	0.9250	0.9500	0.9882	0.9880	0.9633	0.9714
Decision Tree	0.8989	0.8250	0.8675	0.9091	0.9024	0.8806	0.9087
Random Forest	0.9773	0.9268	0.9630	0.9425	0.9639	0.9547	0.9353
AdaBoost	0.9882	0.9157	0.9383	0.9655	0.9630	0.9541	0.9506
Gradient Boosting	0.9535	0.8889	0.9250	0.9425	0.9512	0.9322	0.9481
XGBoost	0.9773	0.9157	0.9383	0.9655	0.9383	0.9470	0.9501

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
Stacking	0.9773	0.9639	0.9756	0.9767	0.9880	0.9763	0.9714

Confusion Matrix & ROC

In [57]:
```
best_model_name = results_df.sort_values("F1 Mean (With-PCA)", ascending=False).ilo
print("\nBest Model (With-PCA):", best_model_name)

best_model = models[best_model_name][0]
best_model.fit(X_pca, y)

y_pred = best_model.predict(X_pca)
cm = confusion_matrix(y, y_pred)
print("\nConfusion Matrix:")
print(cm)

fpr, tpr, _ = roc_curve(y, best_model.predict_proba(X_pca)[:,1])
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr)
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.show()
```

```
Best Model (With-PCA): Logistic Regression

Confusion Matrix:
[[355   2]
 [  6 206]]
```



6. Observations

- Logistic Regression showed clear improvement with PCA.
- Decision Tree and Gradient Boosting improved moderately.
- Naïve Bayes performance decreased after PCA.
- Tree-based ensemble models showed minimal sensitivity to PCA.
- PCA reduced variance across folds for linear models.
- Stacking remained highly robust in both settings.

---

7. Analysis

Linear models benefited from PCA due to:

- Reduction in multicollinearity
- Orthogonal feature representation
- Improved numerical conditioning

Tree-based models showed limited improvement because they inherently perform feature selection.

Naïve Bayes degraded since PCA introduces correlated transformed features, violating its independence assumption.

---

8. Conclusion

PCA was moderately beneficial for this medium-dimensional dataset.

- Significant improvement observed for Logistic Regression.
- Minimal impact on ensemble models.
- Negative impact on Naïve Bayes.
- Stacking remained the best-performing model overall.

Dimensionality reduction is most effective when:

- Feature dimensionality is high
- Multicollinearity is significant
- Overfitting risk is high

For tree-based ensembles, PCA is generally not necessary.

In [ ]: